A Comparative Analysis of Two Capacitor-Less LDO Topologies in 130nm CMOS Technology

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***Abstract*—This paper presents the design and comparison of two capacitor-less (CL) PMOS low dropout (LDO) regulators implemented in a 130nm CMOS technology. Each LDO topology was designed using a different OTA (5T OTA, Telescopic OTA). Both LDO configurations drive a 50 pF load, deliver a load current of 500 µA, support programmable output voltages (2.3 V, 2.1 V and 1.6 V), and have a consumption strictly below 100 µA. Simulation results demonstrate that the 5T OTA provides the most balanced final results, whereas the Telescopic OTA maximizes loop gain at the severe expense of the Error Amplifier's output voltage swing.**

I. Introduction and Theoretical Framework

Low dropout (LDO) voltage regulators are one of the most essential building blocks in power-management systems. Specially with today's increasing demand for system-on-chip solutions, many mixed-signal systems utilize LDO regulators to deliver local supply voltages with reduced noise.

This work will focus on designing two individual LDO architectures. Both LDO topologies must meet the proposed specifications that are presented in Table I. The general LDO schematic is also shown in Figure 1.

TABLE I
Technical LDO Target Specifications

Parameter	Target Value
Power Supply Voltage (V_{EXT})	3 V
Bias Current (I_{bias})	5 µA
Load Current (I_L)	500 µA
Load Capacitance (C_L)	50 pF
Reference Voltage (V_{REF})	$1.2\,\text{V} \pm 1\,\%$
Output Voltage (V_{OUT})	2.3 V
Programmable Trim Voltages	2.1 V, 1.6 V
Compensation Capacitor (C_c)	< 7 pF
Phase Margin @ No-Load	$> 60°$
Phase Margin @ Full-Load	$> 70°$
Loop Gain @ 1 Hz	> 50 dB
PSRR @ 1 Hz	> 65 dB
PSRR @ 1 MHz	> 2 dB
PSRR @ 10 MHz	> 2 dB
Output-Referred Noise @ 10 Hz	$< -115\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 1 kHz	$< -125\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 10 kHz	$< -134\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 100 kHz	$< -143\,\text{dBV}/\sqrt{\text{Hz}}$
Output-Referred Noise @ 1 MHz	$< -155\,\text{dBV}/\sqrt{\text{Hz}}$

As shown in Figure 1, a PMOS LDO is made of an Error Amplifier (EA), a PMOS Pass Device (PD), two feedback resistors R_1 and R_2 for setting the output voltage and a compensation network consisting of a nulling resistor R_c and a capacitor C_c. In this work only one-stage EAs will be presented and analyzed.

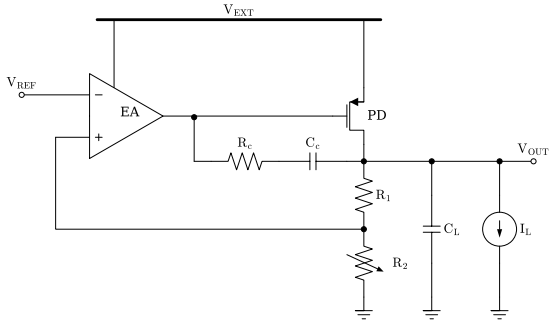


Fig. 1. General LDO Schematic

A. Output Voltage

Assuming that the EA has infinite DC gain (therefore $A_{EA,0} \to \infty$), LDO's gain is given by:

$$\frac{V_{OUT}}{V_{REF}} = \frac{A_{EA,0}}{1 + A_{EA,0}\beta} \xrightarrow{A_{EA,0}\to\infty} \frac{V_{OUT}}{V_{REF}} = \frac{1}{\beta} \tag{1}$$

where β is the feedback factor set by $1 + \frac{R_1}{R_2}$. Therefore:

$$\frac{V_{OUT}}{V_{REF}} = \frac{R_1}{R_1 + R_2} \to \frac{R_1}{R_2} = \frac{V_{REF}}{V_{OUT}} - 1 \tag{2}$$

B. Stability

A small signal model of the LDO schematic can be seen in Figure 2 where the EA's transconductance, output resistance and output load are represented by $g_{m,EA}$, R_1 and C_1 respectively. Similarly, $g_{m,p}$, R_2 and C_2 represent the same values for the PD. The LDO's transfer function has two poles (given that we have a one stage EA), ω_{p1} (dominant) and ω_{p2} (non-dominant) and a zero ω_z, all of which can be approximated with:

$$\omega_{p1} = \frac{1}{g_{m,p} R_1 R_2 C_c} \tag{3}$$

$$\omega_{p2} = \frac{g_{m,p}}{C_1 + C_2} \tag{4}$$

$$\omega_z = \frac{1}{C_c(1/g_{m,p} - R_c)} \tag{5}$$

Even though unity gain bandwidth is not a required specification, one could find the unity gain frequency of the LDO using:

$$UGF \cong \frac{\beta g_{m,EA}}{C_2} \tag{6}$$

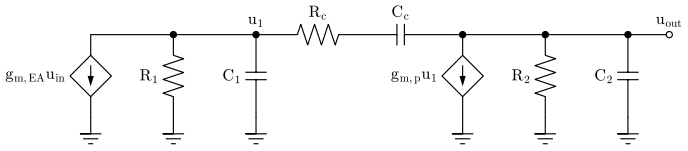


Fig. 2. Small Signal Model of LDO

C. Power Supply Rejection Ratio (PSRR)

Power supply rejection refers to the amount of voltage ripple at the output of the LDO coming from the input voltage and therefore is of great importance. The PSR transfer function of a LDO regulator strongly depends on the type of the EA and whether the input transistors of the EA are NMOS or PMOS (Type-A or Type-B EA respectively). For a PMOS PD, Type-A EAs yield the best PSRR results at low frequencies [2]. Therefore, both LDOs presented in this work utilize Type-A EAs and their PSRR can be approximated by:

$$|PSRR_{DC}| \cong \frac{1}{\beta A_{EA,0} g_{m,p} r_{ds,p}} \tag{7}$$

At higher frequencies, the compensation resistor along with the sizing of the PD can help in achieving the target PSRR requirements.

D. Noise

Noise in LDO regulators refer to the flicker and thermal noise generated by transistors and resistors. Aside from that, the reference voltage can also contribute to the total noise in LDO. If $S_{n,ref}(f)$, $S_{n,EA}(f)$, $S_{n,PD}(f)$, $S_{n,R1}(f)$ and $S_{n,R2}(f)$ represent the noise power spectral density of the voltage reference, error amplifier, pass device and feedback resistors respectively, then the total output noise power spectral density of the LDO regulator is [2]:

$$\begin{aligned} S_{n,o}(f) \cong (S_{n,ref}(f) + S_{n,EA}(f)) \left(1 + \frac{R_1}{R_2}\right)^2 \\ + S_{n,R2}(f)(\frac{R_1}{R_2})^2 + S_{n,R_1}(f) \end{aligned} \tag{8}$$

The EA and feedback resistors noise are typically the dominant sources of a LDO regulator noise.

II. 5T OTA-Based LDO

The 5-transistor OTA used as EA in this LDO variation can be seen in Figure 3. 5T OTA is a simple differential pair with a gain of:

$$A_{5T} = \frac{g_{m,12}}{g_{ds,2} + g_{ds,4}} \tag{9}$$

whereas the output impedance:

$$g_{5T,out} = g_{ds,2} + g_{ds,4} \tag{10}$$

as for the output swing, in order to keep all the transistors in saturation, we must have:

$$V_{REF} - V_T \leq V_{OUT} \leq V_{EXT} - V_{DSAT} \tag{11}$$

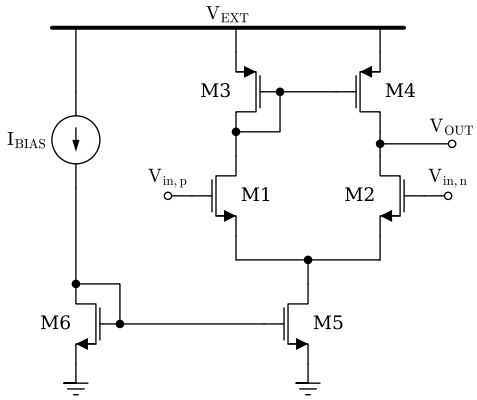


Fig. 3. 5T OTA

TABLE II
Device Sizing for the 5T OTA LDO

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	$42.0/5.0$	µm
M_3, M_4 (Active Load)	$3.8/5.0$	µm
M_5 (Tail Current)	$12.0/5.0$	µm
M_6 (Input Current)	$12.0/5.0$	µm
M_{PD} (Pass Device)	$5.0/5.0$	µm
R_1 (Divider)	15.00	kΩ
R_{21} (Divider)	16.36	kΩ
R_{22} (Divider)	20.00	kΩ
R_{23} (Divider)	45.00	kΩ
R_c (Compensation)	15.00	kΩ
C_c (Compensation)	7.00	pF

In order to satisfy both the PSRR and output voltages, the EA has to deliver enough gain. This can be achieved by increasing the transconductance of the NMOS input pair and the output resistance. Large transistors will also reduce the Flicker noise (as we see later) leading to reduced output referred noise at lower frequencies. Table II shows the device sizing for this LDO variation.

To verify the fundamental operation of the topology, the DC regulation capabilities are evaluated first. Figure 4 illustrates the LDO's ramp response where we can see that the system settles at the target voltages of $2.3\,\mathrm{V}$, $2.1\,\mathrm{V}$, and $1.6\,\mathrm{V}$ with a minimal steady-state offset. This precise tracking confirms that the error amplifier possesses sufficient DC loop gain to minimize steady-state error. Both the loop gain and the loop phase are presented in Figure 6 which also shows the system's phase margin under load. By changing the value of the nulling resistor and therefore the system's zero, the phase margin can either be decreased or increased.

Figure 6 shows the PSRR of the proposed LDO. Simulations prove that the width of the PMOS active load can directly affect the PSRR of LDO and as a result, a small PMOS active load was chosen. The nulling resistor and the sizing of the PD also help in achieving the PSRR at higher frequencies.

As for the output referred noise, as it was mentioned before, larger input transistors help in reducing the DC noise. For noise at higher frequencies, one must increase the current of the transistors, leading to a reduced transistor Thermal noise.

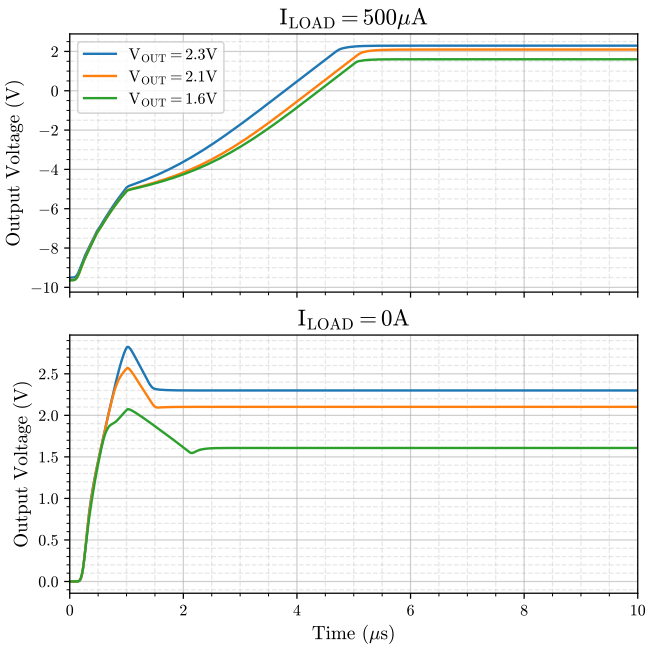


Fig. 4. 5T OTA-Based LDO's Ramp Response

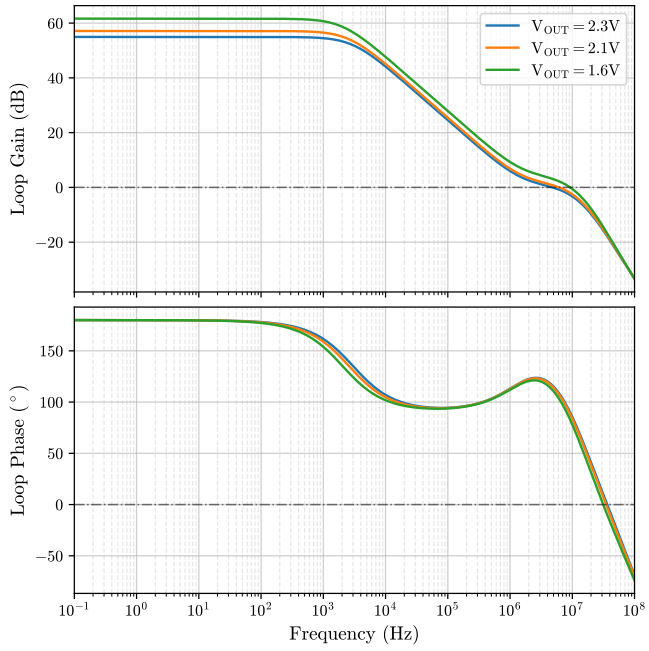


Fig. 5. 5T OTA-Based LDO's Bode Plot Under Load

III. Telescopic OTA-Based LDO

Figure 8 shows the schematic for the Telescopic OTA used in this LDO variation. This OTA has a cascode gain stage with a high-swing current mirror and a self bias network. Therefore, the loop gain and the output impedance are both increased substantially compared to that of the 5T-OTA. More specifically,

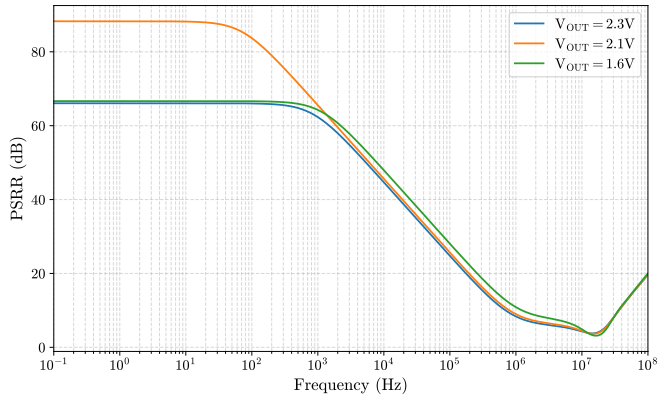


Fig. 6. 5T OTA-Based LDO's PSRR

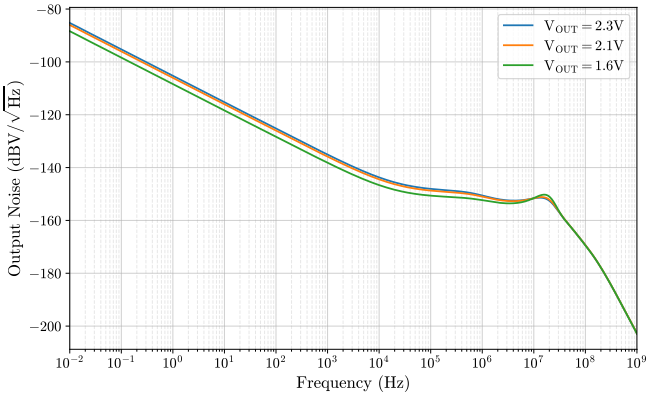


Fig. 7. 5T OTA-Based LDO's Output Referred Noise

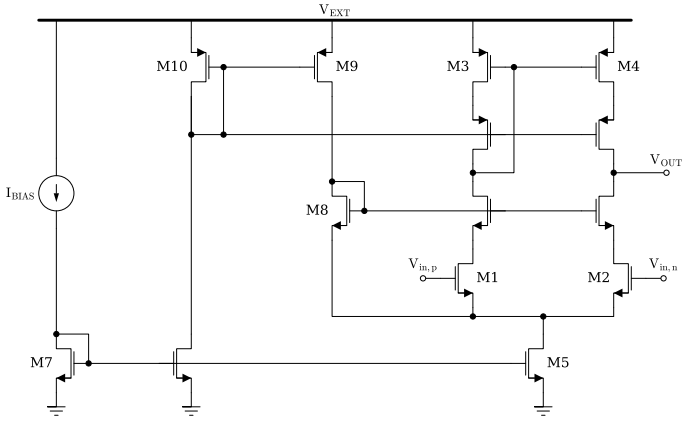


Fig. 8. Telescopic OTA

the loop gain can be given by:

$$A_{TEL} = \frac{g_{m12}}{g_{tel,out}} \tag{12}$$

whereas the output impedance is given by:

$$g_{tel,out} = \frac{1}{g_{ds2}\frac{g_{ds2C}}{g_{m2C}} + g_{ds4}\frac{g_{ds4C}}{g_{m4C}}} \tag{13}$$

Note that M2C and M4C are NMOS and PMOS cascode transistors respectively.

Due to the stacked cascode transistors, the output swing is much more limited in this OTA. In order to ensure that the OTA has the maximum output swing, the bias transistors should be sized using the following equations:

$$\frac{W_8}{L_8} = \frac{W_{3,4}}{4L_{3,4}} \tag{14}$$

$$\frac{W_{10}}{L_{10}} = \frac{W_{1,2}}{4L_{1,2}} \tag{15}$$

Leading to the following output swing:

$$V_{REF} - V_T + V_{DSAT} \leq V_{OUT} \leq V_{EXT} - 2V_{DSAT} \tag{16}$$

Table III contains the sizing parameters. Note that when compared to the previous LDO variation, here we have a smaller bias transistors, bigger feedback resistors and larger PD width. This happens to ensure that the gate voltage of the PD is well within the output range of the Telescopic OTA.

The ramp response can be seen in Figure 9 where all three target output voltages are achieved. Furthermore, Figure 10 presents the bode plot for this LDO variation. One can see that the gain is much larger compared to that of the 5T-OTA and therefore we expect to get better PSRR results.

TABLE III
DEVICE SIZING FOR THE TELESCOPIC OTA LDO

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	25.0/5.0	µm
M_{1C}, M_{2C} (NMOS Cascode)	0.65/1.0	µm
M_{3C}, M_{4C} (PMOS Cascode)	2.0/1.0	µm
M_3, M_4 (Active Load)	40.0/5.0	µm
M_8 (NMOS Bias)	0.65/4.0	µm
M_9, M_{10} (PMOS Bias)	2.0/4.0	µm
M_5 (Tail Current)	0.95/1.0	µm
M_6 (Bias Current)	0.28/1.0	µm
M_7 (Input Current)	0.28/1.0	µm
M_{PD} (Pass Device)	40.0/0.13	µm
R_1 (Divider)	22.00	kΩ
R_{21} (Divider)	24.00	kΩ
R_{22} (Divider)	29.33	kΩ
R_{23} (Divider)	66.00	kΩ
R_c (Compensation)	20.00	kΩ
C_c (Compensation)	7.00	pF

Figure 11 illustrates the PSRR results where we can see are better compared to that of the 5T-OTA due to higher loop gain. Figure 12 visualizes the output referred noise which is identical to the one of the 5T-OTA, mostly because the cascode transistors don't have much noise contribution.

IV. COMPARISON

Table IV displays the performance of both LDO variation in addition to the target specifications. Both LDOs presented in this work satisfy all target specifications except the specification for output-referred noise @ 1 MHz. 5T OTA should be preferred when smaller PD is required or when one needs more output swing. Telescopic OTA, on the other hand, maximizes gain and PSRR at the severe expense of output swing.

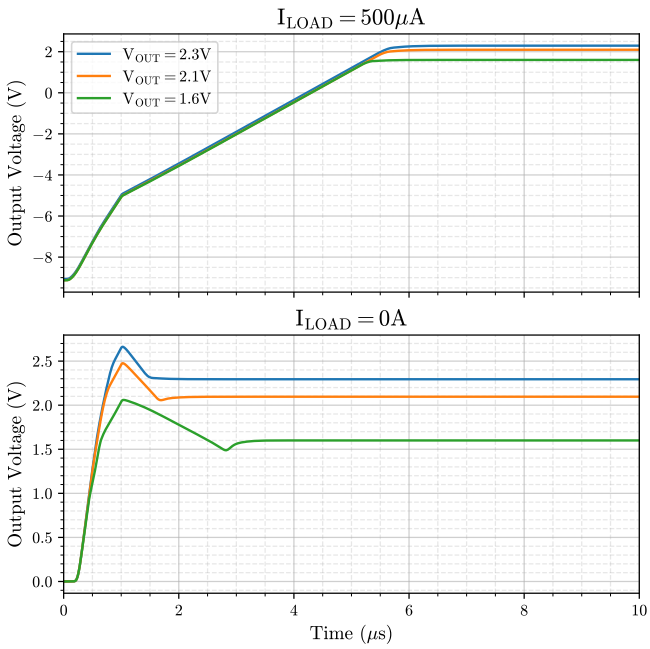


Fig. 9. Telescopic OTA-Based LDO's Ramp Response

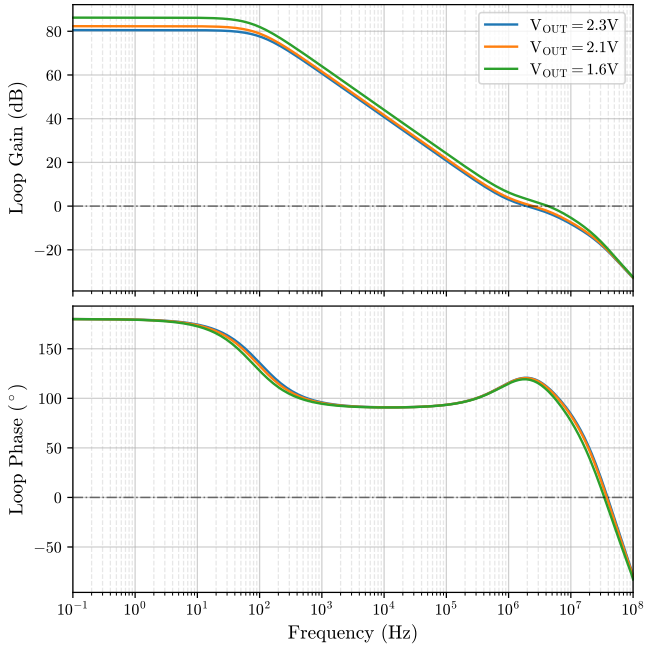


Fig. 10. Telescopic OTA-Based LDO's Bode Plot Under Load

V. FUTURE WORK

In order to achieve the noise specification at higher frequencies, one must increase the bias current of the EA's input transistors, given that at higher frequencies it is the transistors' Thermal noise that dominates. Unfortunately, given that for both displayed topologies the increase of bias current decreases the output impedance, we will witness lower loop gain and therefore lower PSRR values. Therefore, other topologies can be designed

TABLE IV
SIMULATED PERFORMANCE SUMMARY (ACROSS ALL TRIMS)

Parameter	**Target Specification**	**5T-OTA (Min – Max)**	**Telescopic-OTA (Min – Max)**
Trim Voltage 1	$= 2.3\,\mathrm{V}$	2.293 – 2.299	2.294
Trim Voltage 2	$= 2.1\,\mathrm{V}$	2.096 – 2.102	2.096
Trim Voltage 3	$= 1.6\,\mathrm{V}$	1.602 – 1.607	1.6
Quiescent Current	$< 100\,\mu\mathrm{A}$	48.16 – 94.49	52.56 – 84.12
Phase Margin ($I_{load} = 0$)	$> 60°$	77.62 – 99.2	74.15 – 99
Phase Margin ($I_{load} = 500\,\mu\mathrm{A}$)	$> 70°$	82.7 – 116.2	107.2 – 120.1
Loop Gain @ 1 Hz	$> 50\,\mathrm{dB}$	54.95 – 61.63	80.53 – 86.23
PSRR @ 1 Hz	$> 65\,\mathrm{dB}$	66.07 – 88.27	70.73 – 75.86
PSRR @ 1 MHz	$> 2\,\mathrm{dB}$	8.43 – 10.95	6.373 – 8.537
PSRR @ 10 MHz	$> 2\,\mathrm{dB}$	4.353 – 4.934	2.107 – 2.118
Output-Referred Noise @ 10 Hz	$< -115\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$	-118.3 – -115.1	-118.6 – -115.5
Output-Referred Noise @ 1 kHz	$< -125\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$	-138.1 – -135	-138.1 – -135.1
Output-Referred Noise @ 10 kHz	$< -134\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$	-146.5 – -143.6	-145.4 – -142.5
Output-Referred Noise @ 100 kHz	$< -143\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$	-150.6 – -148	-148 – -145.3
Output-Referred Noise @ 1 MHz	$< -155\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$	-152.3 – -150.5	-149.9 – -148.3

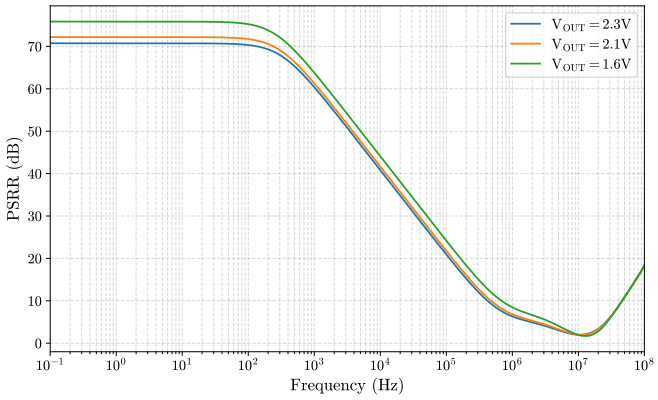


Fig. 11. Telescopic OTA-Based LDO's PSRR

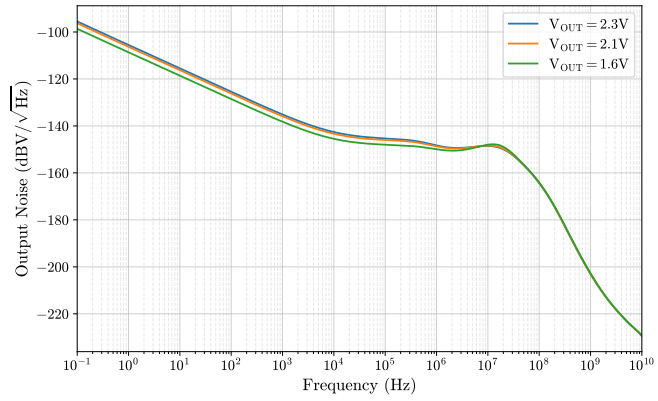


Fig. 12. Telescopic OTA-Based LDO's Output Referred Noise

and tested where there are individual input and gain stages with different currents. One such topology would be the Current Mirror OTA displayed in Figure 13.

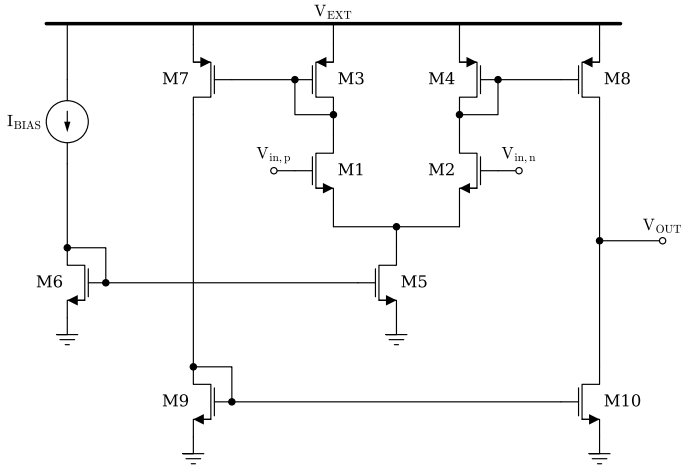


Fig. 13. Current Mirror OTA-Based LDO

ACKNOWLEDGMENTS

This work wouldn't have been possible without the help of PhD candidate Andreas Papathanasiou and Pr. Paul-Peter Sotiriadis. Special thanks go to Dr. Dimitrios Baxevanakis for his guidance and help all throughout the project.

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